DocuMind AI Architecture Whitepaper

Evidence-Grounded RAG & Claim Verification Systems

1. ARCHITECTURAL OVERVIEW

DocuMind AI is an enterprise document intelligence platform engineered for factual precision, verifiable citations, and automated contradiction detection across enterprise document repositories.

2. HYBRID RETRIEVAL PIPELINE

To solve common lexical and semantic search failure modes, DocuMind integrates a multi-stage hybrid retrieval architecture:
- Dense Vector Retrieval: Employs the sentence-transformers 'all-MiniLM-L6-v2' model, generating 384-dimensional dense semantic embeddings with cosine distance metric.
- Sparse Keyword Retrieval: Utilizes BM25 (Best Matching 25) indexing with calibrated hyperparameters k1 = 1.5 and b = 0.75.
- Reciprocal Rank Fusion (RRF): Combines dense and sparse rank lists using the formula:
  RRF_Score(d) = Sum_{m in M} (1 / (k + rank_m(d)))
  where default fusion constant k = 60.
- Deduplication and Chunk Preservation: Preserves page numbers, character spans, and document IDs across all retrieved segments.

DocuMind AI Architecture Whitepaper

Cross-Encoder Reranking and Claim Verification

3. CROSS-ENCODER RERANKING

Top-30 candidate passages from RRF fusion are passed through an optional cross-encoder reranking model:
- Model Architecture: ms-marco-MiniLM-L-6-v2.
- Scoring: Computes joint query-passage cross-attention scores to refine the top-5 to top-10 context passages passed to the generator.

4. VECTOR STORAGE CONFIGURATION

- Vector Engine: ChromaDB persistent vector database.
- Similarity Metric: Cosine distance (1 - cosine_similarity).
- Sharding & Metadata: Page number, chunk index, document hash, and collection ID stored in indexed metadata attributes.

5. EVIDENCE VERIFICATION ENGINE

The verification engine decomposes generated responses into atomic factual claims. Each claim is evaluated against source passages:
- Supported: Evidence directly entails the claim.
- Partially Supported: Evidence supports part of the claim or introduces minor qualification.
- Contradicted: Evidence directly refutes or contradicts the claim.
- Not Supported: Evidence does not contain sufficient information to prove the claim.

6. CROSS-DOCUMENT CONFLICT DETECTION

The engine performs pairwise statement comparison across multi-version documents to flag divergent dates, numbers, and policies.